Structural Load Analysis of a Simply Supported Beam

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Title

Structural Load Analysis of a Simply Supported Reinforced Concrete Beam

Introduction

Structural load analysis is an essential part of civil engineering that helps determine the strength, stability, and safety of structural members. Structures are subjected to different types of loads during their service life, such as dead loads, live loads, and environmental loads like wind and earthquakes. Proper load analysis ensures that structures can safely withstand these forces without failure.

Beam Specifications

Parameter	Value
Beam Type	Simply Supported Beam
Material	Reinforced Concrete
Length	4 m
Width	0.3 m
Depth	0.5 m
Unit Weight of Concrete	25 kN/m^3

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Dead Load Calculation

Dead load refers to the self-weight of the structural member.

Formula:

Dead Load = Unit Weight × Volume

Volume of Beam = Length × Width × Depth

$= 4 \times 0.3 \times 0.5$

$= 0.6\ m^3$

Dead Load = 25×0.6

= **15 kN**

Live Load Calculation

Live load represents the movable or temporary loads acting on the structure.

Assumed Live Load = $3\ kN/m^2$

Area = Length × Width

$= 4 \times 0.3$

$= 1.2\ m^2$

Live Load = 3×1.2

= **3.6 kN**

Wind Load Calculation

For basic analysis, the wind load is assumed to be:

Wind Load = **1.5 kN**

Total Load Calculation

Total Load = Dead Load + Live Load + Wind Load

$= 15 + 3.6 + 1.5$

= **20.1 kN**

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Structural Behavior

The beam experiences bending due to the applied loads. The maximum bending moment occurs near the center of the beam, while maximum shear forces occur near the supports. As the load increases, the deflection of the beam also increases. Proper structural design ensures that the beam remains stable and safe under these loading conditions.

Conclusion

The analysis shows that the simply supported reinforced concrete beam is subjected to a total load of **20.1 kN**. Considering dead load, live load, and wind load, the beam is capable of safely resisting the applied forces. Structural load analysis plays a significant role in ensuring the safety, durability, and performance of structures.

References

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4. S. Unnikrishna Pillai and Devdas Menon, *Reinforced Concrete Design*.
5. B.C. Punmia, *Strength of Materials and Theory of Structures*.